

File Edit Selection View Go Run Terminal Help Python\_Test

EXPLORER

- PYTHON\_TEST
  - color\_splash.png
  - colorspaces.py
  - composite\_masterpiece.png
  - create\_challenge\_images.py
  - create\_test\_image\_day3.py
  - create\_test\_image.py
  - crop.py
  - custom\_cyberpunk.png
  - custom\_mask.png
  - custom\_sketch.png
  - custom\_vintage.png
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  - day6\_arithmetic\_bitwise.py
  - day7\_mini\_challenge.py
  - day8\_colour\_spaces.py
  - day9\_histogram.py
  - dissolve.mp4
  - drawin\_basics.py
  - drawing.py
  - ex\_1\_day\_3.py
  - ex\_2\_day\_3.py
  - ex\_3\_day\_3.py
  - ex\_4\_day\_3.py
  - exercise\_1\_custom\_pipeline.py
  - exercise\_1\_house.py
  - exercise\_1.py
  - exercise\_2\_comparison.py
  - exercise\_2\_target.py
  - exercise\_2.py
  - exercise\_3\_chessboard.py
  - exercise\_3.py

day9\_histogram.py

```
243 # Apply to real image
244 print("\n--- Applying to your image ---")
245 equalized_real = cv2.equalizeHist(image_gray)
246 # Show comparison
247 comparison = np.hstack([image_gray, equalized_real])
248 cv2.imshow("Original vs Equalized (Left: Original, Right: Equalized)", comparison)
249 cv2.waitKey(0)
250 # Save result
251 cv2.imwrite("equalized_image.png", equalized_real)
252 print("Saved: equalized_image.png")
253 print("\nNote: Histogram equalization works best on images with poor contrast.
254       - The image has poor contrast
255       - Foreground and background are both dark
256       - Medical and satellite images (can be improved with histogram equalization)")
257 # SECTION 7: HISTOGRAMS WITH MASKS
258 # -----
259 print("\n[Section 7] Histograms with Masks")
260 print("-" * 50)
261 print("Masks allow us to compute histograms for specific regions of an image.")
262 # Create a mask (circular region in center)
263 mask = np.zeros(image_gray.shape[:2], dtype=bool)
264 center_x, center_y = width // 2, height // 2
265 radius = 100
```

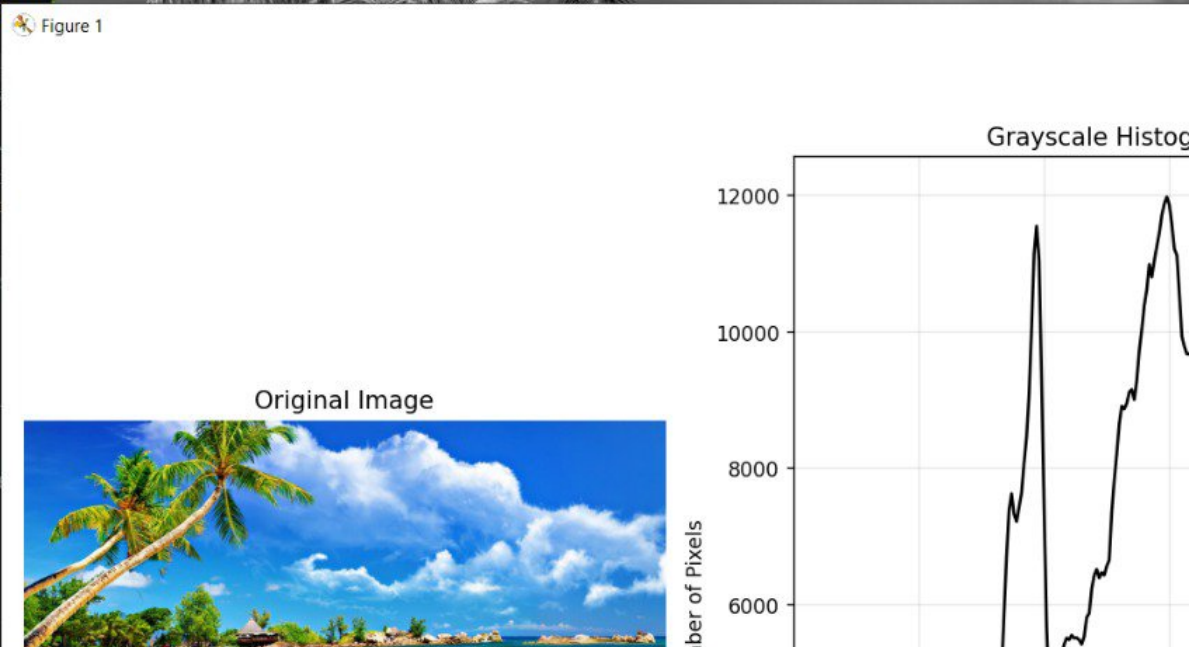
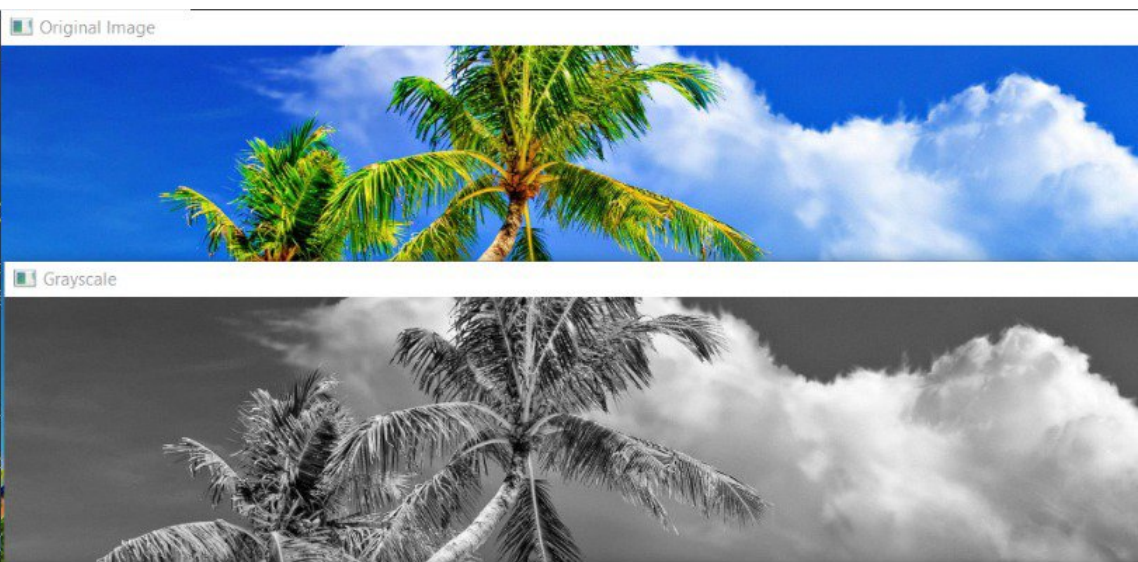
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORT

Loaded: 1437 x 826 pixels

[Section 2] Computing Grayscale Histogram...

Method 1: cv2.calcHist()  
Histogram shape: (256, 1)  
Total pixels counted: 1186962.0 (should equal 1186962)

Method 2: NumPy histogram  
NumPy histogram shape: (256,)  
Methods match: True



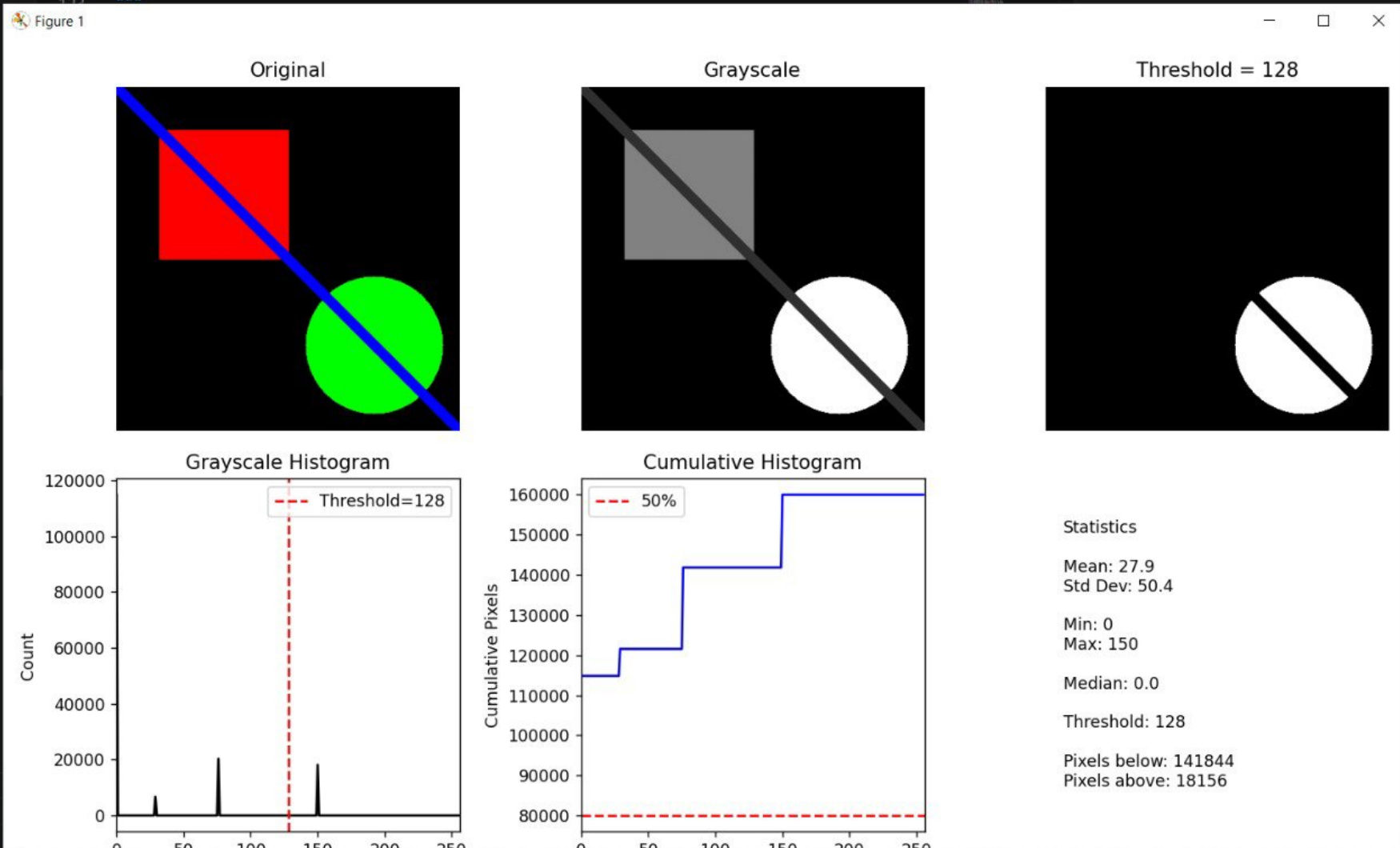
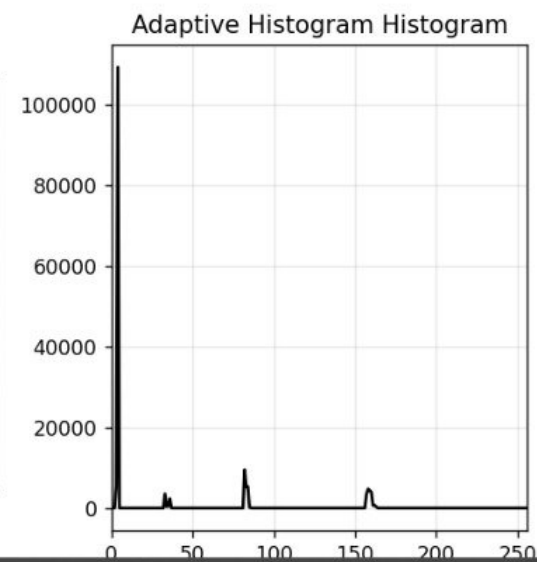
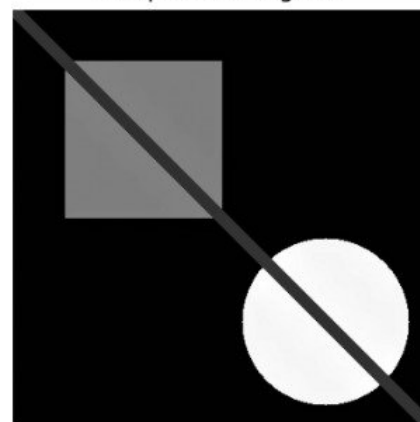
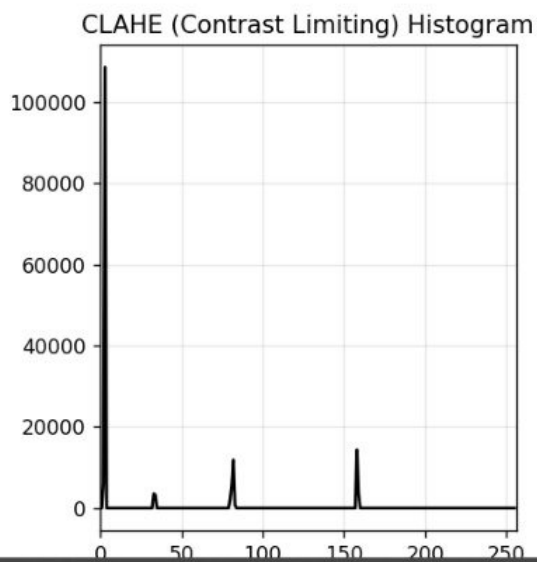
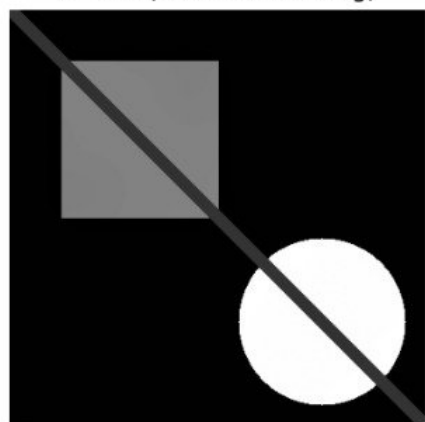
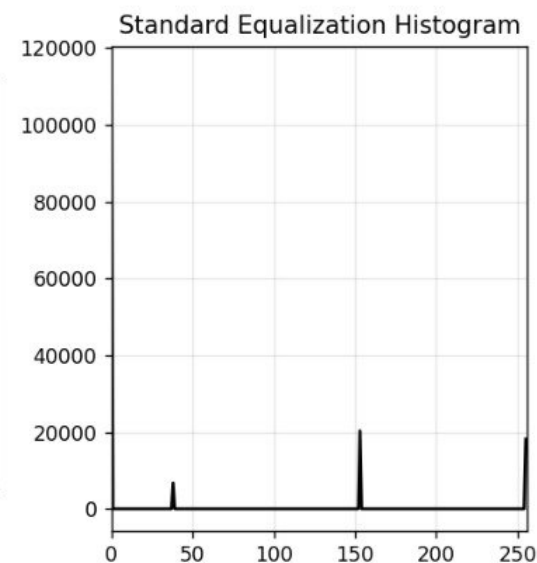
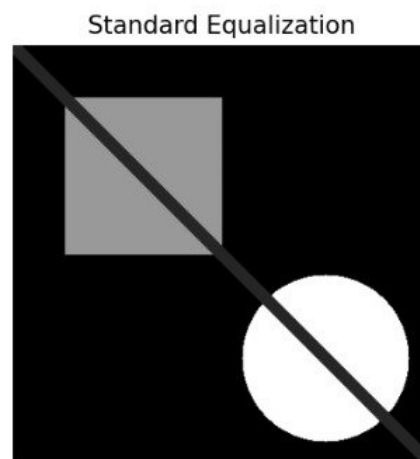
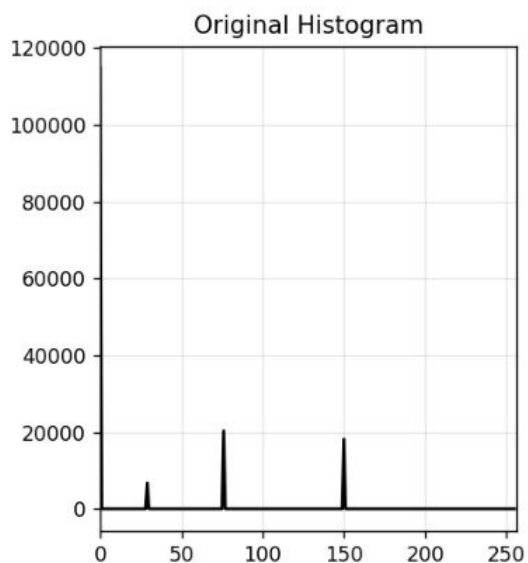
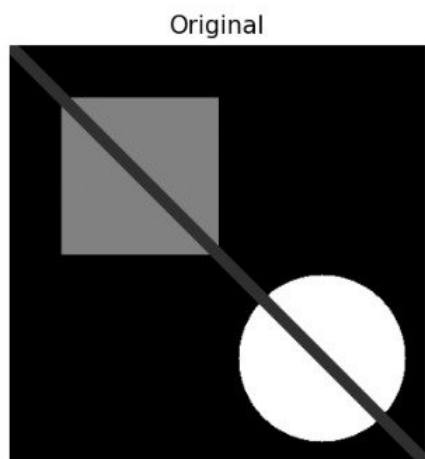
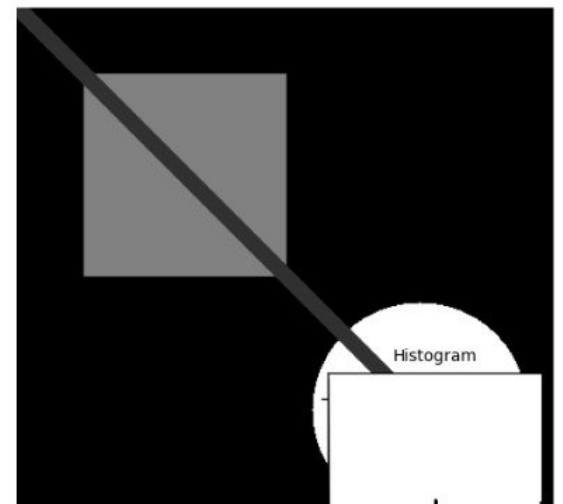
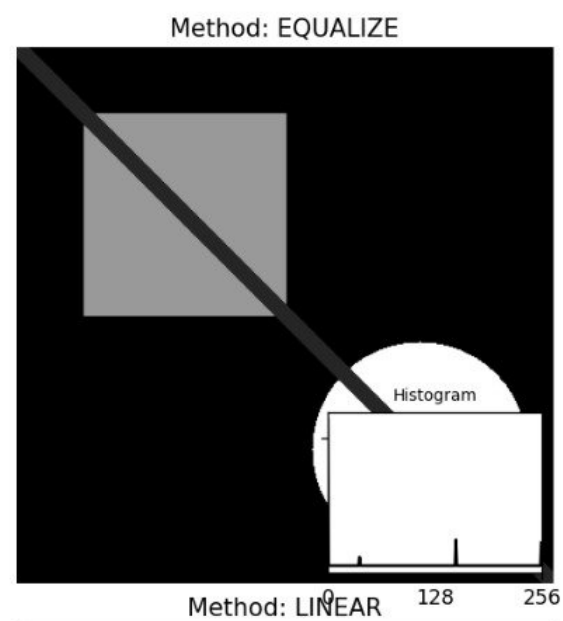




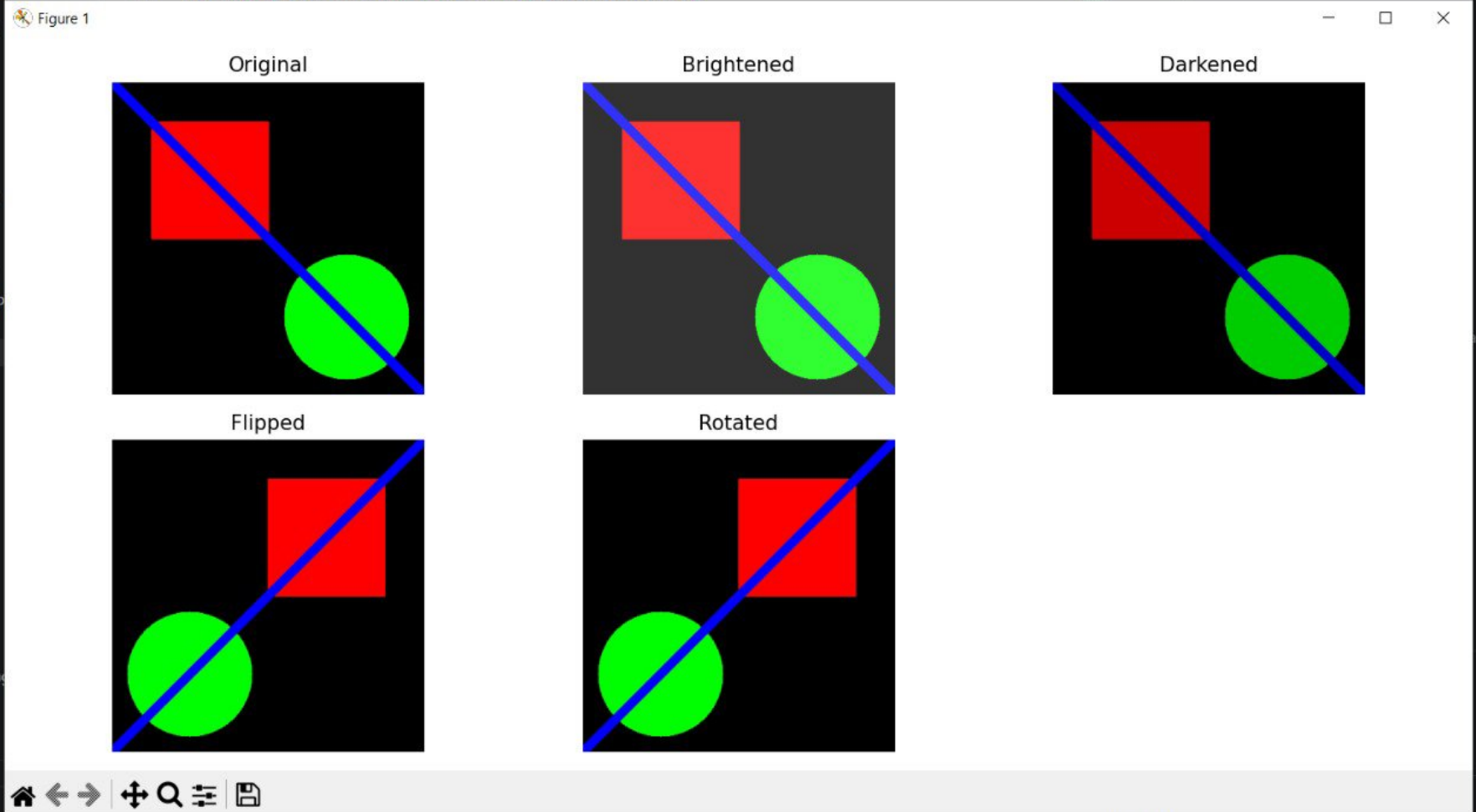
Figure 1



— □ ×



```
87  
88 for idx, (name, variant) in enumerate(variants.items()):
```



Python\_Test

exercise3\_3d\_color\_histogram.py

25 plt.savefig("3d\_color\_distribution.png")

26 plt.show()

27 # Create

28 fig, axes

29 projection

30 (0, 1, "

31 (0, 2, "

32 (1, 2, "

33 ]

34 for idx,

35 axes[

36 c=sam

37 axes[

38 axes[

39 axes[

40 axes[

41 axes[

42 plt.tight

43 plt.savefig

44 plt.show()

Figure 1

### 3D RGB Color Space Distribution

The figure is a 3D plot titled "3D RGB Color Space Distribution". It shows a cube representing the RGB color space. The axes are labeled "Red", "Green", and "Blue". The "Red" axis ranges from 0 to 250, the "Green" axis ranges from 0 to 250, and the "Blue" axis ranges from 0 to 250. A small black dot is visible on the Blue axis near the origin, and a small red dot is visible on the Red axis near the origin.

PS C:\Users\jebro

Type here to search

34°C Partly sunny

17:13 12-06-2026